

Ph134, Lecture 2

- **Properties of a continuous probability distribution**
- **Warmup: error propagation**
- **Given a “true” single value μ with measurement results subject to random error, Use N trials to estimate the value of μ :**
 - **Mean and sample variance**
 - **Standard error of the mean (SEM)**

Properties of a continuous probability

distribution: mean μ , variance σ^2

$$\int_{-\infty}^{\infty} P(y) dy = 1$$

Note $P(y)$ is not a probability, but a probability density:

$P(y)dy$ is the probability of getting a measurement result between y and $y+dy$.

Mean:

$$\langle y \rangle \equiv \int_{-\infty}^{\infty} y P(y) dy = \mu$$

Variance:

$$\begin{aligned} \langle (y - \mu)^2 \rangle \\ \equiv \int_{-\infty}^{\infty} (y - \mu)^2 P(y) dy = \sigma^2 \end{aligned}$$

Standard deviation: $\sqrt{\sigma^2} = \sigma$

For a normal distribution:

$$\int_{\mu-\sigma}^{\mu+\sigma} P(y) dy = 0.68$$

$$\int_{\mu-2\sigma}^{\mu+2\sigma} P(y) dy = 0.95$$

$$\int_{\mu-3\sigma}^{\mu+3\sigma} P(y) dy = 0.997$$

Warmup: Error propagation

y drawn from $P(y)$ with mean μ_y , variance σ_y^2

z drawn from $P(z)$ with mean μ_z , variance σ_z^2

y and *z* drawn independently, so $P(y,z)=P(y)P(z)$

You're interested in some function $f(y,z)$.

Suppose a linear approximation is justified:

$$f(y,z) \approx f(\mu_y, \mu_z) + \left. \frac{\partial f}{\partial y} \right|_{\mu_y, \mu_z} (y - \mu_y) + \left. \frac{\partial f}{\partial z} \right|_{\mu_y, \mu_z} (z - \mu_z)$$

Then:

$$\left\langle (f(y,z) - f(\mu_y, \mu_z))^2 \right\rangle \approx \left(\frac{\partial f}{\partial y} \right)^2 \left\langle (y - \mu_y)^2 \right\rangle + \left(\frac{\partial f}{\partial z} \right)^2 \left\langle (z - \mu_z)^2 \right\rangle$$

$$\sigma_f^2 \approx \left(\frac{\partial f}{\partial y} \right)^2 \sigma_y^2 + \left(\frac{\partial f}{\partial z} \right)^2 \sigma_z^2$$

From N trials to a value and an uncertainty:

Task: Given N independent trials $\{y_1, y_2, \dots, y_N\}$, i.e., N values sampled at random from the distribution $P(y)$ with mean μ and variance σ^2 unknown to us, estimate μ and attach an uncertainty to that estimate of μ .

Claimed solution (we will prove):

Sample mean

$$\bar{y} = \frac{1}{N} \sum_{n=1}^N y_n \quad \text{estimates } \mu.$$

Sample variance

$$s^2 = \frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2 \quad \text{estimates } \sigma^2.$$

Standard error of the mean
(SEM)

$$s_{\bar{y}} = \frac{s}{\sqrt{N}} \quad \begin{array}{l} \text{gives uncertainty of } \bar{y} \\ \text{as an estimate of } \mu. \end{array}$$

Sample mean is an unbiased estimator of μ :

Take N independent trials $\{y_1, y_2, \dots, y_N\}$

Calculate

$$\bar{y} = \frac{1}{N} (y_1 + y_2 + \dots + y_N)$$

$$\bar{y} = \frac{1}{N} \sum_{n=1}^N y_n$$

$$\langle \bar{y} \rangle = \mu$$

What is the variance of $\bar{y}$?

$$\bar{y} = \frac{1}{N} (y_1 + y_2 + \cdots + y_N) = \frac{1}{N} \sum_{n=1}^N y_n$$

$$\langle (y_n - \mu)^2 \rangle = \sigma^2$$

$$\langle (\bar{y} - \mu)^2 \rangle = \frac{\sigma^2}{N}$$

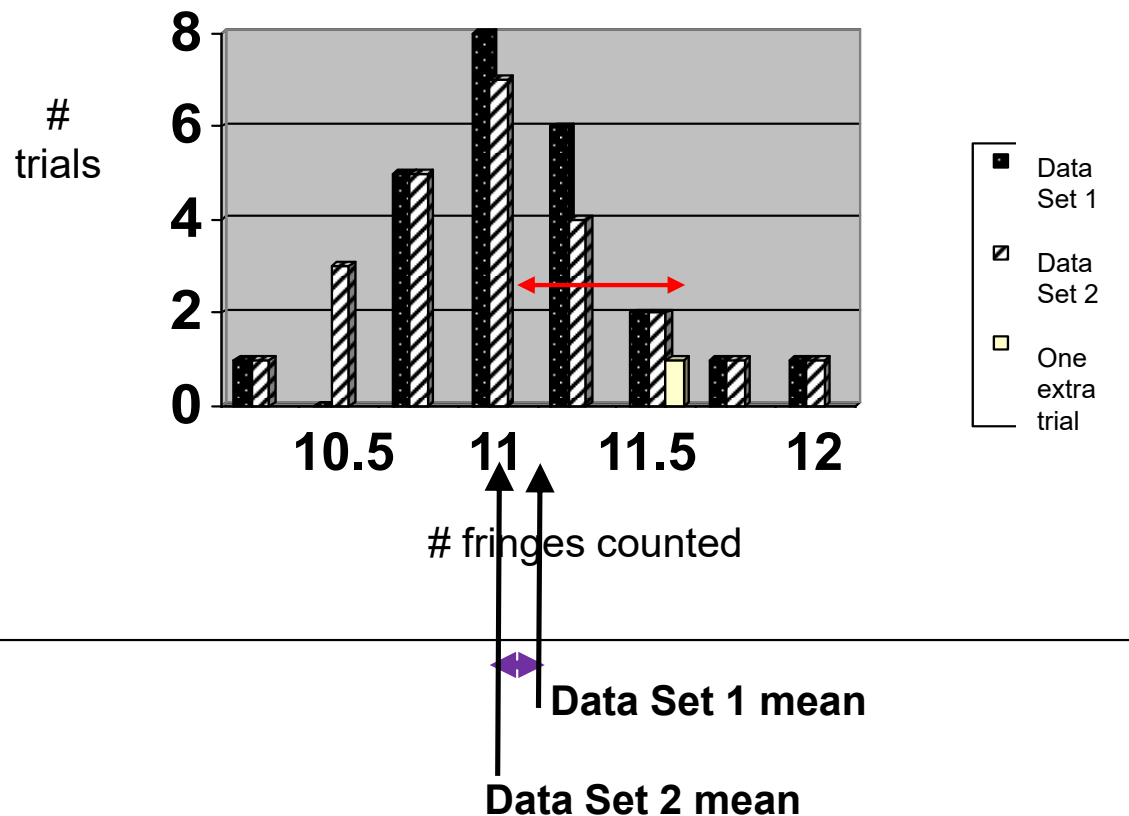
SEM (Standard Error of the Mean)

$$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{N}}$$

So we should use our N trials to say $\mu = \bar{y} \pm \frac{\sigma}{\sqrt{N}}$

Disclaimer: The following data sets are completely made up to illustrate the concept of standard deviation vs. standard error. Do not expect the data of our friend I.N.Competent to resemble your own (real) data, either in spread or in mean value!

Fringes Passing as Pressure Changes
from 50mm to 400mm Hg



Standard deviation:

Comparable to width of either data set, or expected distance of “one extra trial” from the mean of either data set.

Standard error (of the mean):

*Comparable to deviation of either **mean** from the true mean, and thus comparable to the deviation of the Data Set 1 and Data Set 2 means from each other.*

What is the variance of $\bar{y}$?

$$\bar{y} = \frac{1}{N} (y_1 + y_2 + \cdots + y_N) = \frac{1}{N} \sum_{n=1}^N y_n$$

$$\langle (y_n - \mu)^2 \rangle = \sigma^2$$

$$\langle (\bar{y} - \mu)^2 \rangle = \frac{\sigma^2}{N}$$

SEM (Standard Error of the Mean)

$$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{N}}$$

So we should use our N trials to say $\mu = \bar{y} \pm \frac{\sigma}{\sqrt{N}}$

...but we don't know the value of σ .

Sample variance is an unbiased estimator of σ^2 :

Sample variance

$$s^2 = \frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2$$

Claim: $\langle s^2 \rangle = \sigma^2$

Beginning of proof:

$$\begin{aligned} \langle s^2 \rangle &= \left\langle \frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2 \right\rangle \\ &= \frac{1}{N-1} \sum_{n=1}^N \langle (y_n - \bar{y})^2 \rangle \end{aligned}$$

*Need to know what
this is.*

*It's **SMALLER**
THAN σ^2 ,
because y_n helped
determine $\bar{y}$.*

From **N** trials to a value and an uncertainty:

Given N independent trials $\{y_1, y_2, \dots, y_N\}$, i.e., N values sampled at random from the distribution $P(y)$ with true mean μ unknown to us,

we conclude $\mu = \bar{y} \pm \frac{s}{\sqrt{N}}$, where $s = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (y_n - \bar{y})^2}$

Issues to be careful about?

- make the trials as independent as possible! E.g., if measuring $y(x)$, reset x carefully before each trial.*
- accounting for random error in all this, but never for systematic error*

This time: turning N trials into a point with an error bar

Starting next time: plotting a set of points with error bars and fitting a function to them!

